
Stochastic Calculus for Finance II

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1. Probability Preliminaries

1.1 Definitions, Theorems, Corollaries

Definition 1.1.1. Let Ω be a nonempty set, and let $\mathcal{F}$ be a collection of subsets of Ω . $\mathcal{F}$ is a σ -field/algebra if it has:

- (i) Closure under countable unions
- (ii) Closure under countable intersections
- (iii) Closure under complements

Definition 1.2.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable is a real-valued function X defined on Ω with the property that for every Borel subset $\mathcal{B}$ of $\mathbb{R}$, the subset of Ω given by

$$\{X \in \mathcal{B}\} = \{\omega \in \Omega; X(\omega) \in \mathcal{B}\}$$

is in the σ -algebra $\mathcal{F}$. We sometimes permit a random variable to take the values $+\infty$ and $-\infty$. *For example, this pops up when making a change of variables or dealing with a random variable in a trig space (i.e. arctan in $[-\pi/2, \pi/2]$)*

Definition 1.2.3. Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The distribution measure of X is the probability measure μ_X that assigns to each Borel subset $\mathcal{B}$ of $\mathbb{R}$ the mass $\mu_X(\mathcal{B}) = \mathbb{P}\{X \in \mathcal{B}\}$. *We could talk about Dirac delta functions where we assign the density to a point mass of 1 or 0, but otherwise this definition is fairly intuitive from M463.*

Theorem 1.3.1. Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- (i) If X takes only finitely many values $y_0, y_1, y_2, \dots, y_n$, then

$$\int_{\Omega} X(\omega) d\mathbb{P}(\omega) = \sum_{k=0}^n y_k \mathbb{P}\{X = y_k\}.$$

- (ii) **(Integrability)** The random variable X is integrable if and only if

$$\int_{\Omega} |X(\omega)| d\mathbb{P}(\omega) < \infty.$$

Now let Y be another random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

- (iii) **(Comparison)** If $X \leq Y$ almost surely (i.e., $\mathbb{P}\{X \leq Y\} = 1$), and if $\int_{\Omega} X(\omega) d\mathbb{P}(\omega)$ and $\int_{\Omega} Y(\omega) d\mathbb{P}(\omega)$ are defined, then

$$\int_{\Omega} X(\omega) d\mathbb{P}(\omega) \leq \int_{\Omega} Y(\omega) d\mathbb{P}(\omega).$$

In particular, if $X = Y$ almost surely and one of the integrals is defined, then they are both defined and their expectations are equal.

(iv) **(Linearity)** If α and β are real constants and X and Y are integrable, or if α and β are nonnegative constants and X and Y are nonnegative, then

$$\int_{\Omega} (\alpha X(\omega) + \beta Y(\omega)) d\mathbb{P}(\omega) = \alpha \int_{\Omega} X(\omega) d\mathbb{P}(\omega) + \beta \int_{\Omega} Y(\omega) d\mathbb{P}(\omega).$$

Jensen's Inequality. If ϕ is a convex, real-valued function defined on $\mathbb{R}$, and if $\mathbb{E}[|X|] < \infty$, then

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)].$$

Definition 1.3.7. Let $\mathcal{B}(\mathbb{R})$ be the σ -algebra of Borel subsets of $\mathbb{R}$. The Lebesgue measure on $\mathbb{R}$, denoted $\mathcal{L}$, assigns to each set $B \in \mathcal{B}(\mathbb{R})$ a measure satisfying the countable additivity property

$$\mathcal{L}\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mathcal{L}(B_n)$$

for pairwise disjoint sets $B_1, B_2, \dots \in \mathcal{B}(\mathbb{R})$.

Theorem 1.3.8. (Comparison of Riemann and Lebesgue Integrals). Let f be a bounded function defined on $\mathbb{R}$, and let $a < b$ be numbers.

(i) The Riemann integral $\int_a^b f(x) dx$ is defined (i.e., the lower and upper Riemann sums converge to the same limit) if and only if the set of points x in $[a, b]$ where $f(x)$ is not continuous has Lebesgue measure zero.

(ii) If the Riemann integral $\int_a^b f(x) dx$ is defined, then f is Borel measurable (so the Lebesgue integral $\int_{[a,b]} f(x) d\mathcal{L}(x)$ is also defined), and the Riemann and Lebesgue integrals agree.

Definition 1.3.9. If the set of numbers in $\mathbb{R}$ that fail to have some property is a set with Lebesgue measure zero, we say that the property holds *almost everywhere*.

Definition 1.4.1. Let $X_1, X_2, X_3, \dots$ be a sequence of random variables, all defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X be another random variable defined on this space. We say that $X_1, X_2, X_3, \dots$ converges to X *almost surely* and write

$$\lim_{n \rightarrow \infty} X_n = X \text{ almost surely}$$

if the set of $\omega \in \Omega$ for which the sequence of numbers $X_1(\omega), X_2(\omega), X_3(\omega), \dots$ has limit $X(\omega)$ is a set with probability one. Equivalently, the set of $\omega \in \Omega$ for which $X_1(\omega), X_2(\omega), X_3(\omega), \dots$ does not converge to $X(\omega)$ has probability zero.

Definition 1.4.3. Let $f_1, f_2, f_3, \dots$ be a sequence of real-valued, Borel-measurable functions defined on $\mathbb{R}$. Let f be another real-valued, Borel-measurable function defined on $\mathbb{R}$. We say that $f_1, f_2, f_3, \dots$ converges to f *almost everywhere* and write

$$\lim_{n \rightarrow \infty} f_n = f \text{ almost everywhere}$$

if the set of $x \in \mathbb{R}$ for which the sequence of numbers $f_1(x), f_2(x), f_3(x), \dots$ does not have limit $f(x)$ is a set with Lebesgue measure zero.

Theorem 1.4.5 (Monotone Convergence). Let $X_1, X_2, X_3, \dots$ be a sequence of random variables converging almost surely to another random variable X . If

$$0 \leq X_1 \leq X_2 \leq X_3 \leq \dots \text{ almost surely,}$$

then

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Let $f_1, f_2, f_3, \dots$ be a sequence of Borel-measurable functions on $\mathbb{R}$ converging almost everywhere to a function f . If

$$0 \leq f_1 \leq f_2 \leq f_3 \leq \dots \text{ almost everywhere,}$$

then

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx = \int_{-\infty}^{\infty} f(x) dx.$$

Corollary 1.4.6. Suppose the nonnegative random variable X takes countably many values $x_0, x_1, x_2, \dots$. Then

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} x_k \mathbb{P}\{X = x_k\}.$$

Theorem 1.4.9 (Dominated Convergence). Let $X_1, X_2, \dots$ be a sequence of random variables converging almost surely to a random variable X . If there is another random variable Y such that $\mathbb{E}[Y] < \infty$ and $|X_n| \leq Y$ almost surely for every n , then

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Intuitive example: $\text{Bin}(n, \lambda/n) \Rightarrow \text{Pois}(\lambda)$.

Let $f_1, f_2, \dots$ be a sequence of Borel-measurable functions on $\mathbb{R}$ converging almost everywhere to a function f . If there is another function g such that $\int_{-\infty}^{\infty} g(x) dx < \infty$ and $|f_n| \leq g$ almost everywhere for every n , then

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx = \int_{-\infty}^{\infty} f(x) dx.$$

Theorem 1.5.1. Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let g be a Borel-measurable function on $\mathbb{R}$. Then

$$\mathbb{E}[|g(X)|] = \int_{\mathbb{R}} |g(x)| d\mu_X(x),$$

and if this quantity is finite, then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) d\mu_X(x).$$

Theorem 1.5.2. Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let g be a Borel-measurable function on $\mathbb{R}$. Suppose that X has a density f (i.e., f is a function satisfying $\mu_X(B) = \int_B f(x) dx$ for all Borel subsets $B \subset \mathbb{R}$). Then

$$\mathbb{E}[|g(X)|] = \int_{-\infty}^{\infty} |g(x)| f(x) dx.$$

If this quantity is finite, then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx.$$

Theorem 1.6.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let Z be an almost surely non-negative random variable with $\mathbb{E}[Z] = 1$. For $A \in \mathcal{F}$, define

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega).$$

Then $\tilde{\mathbb{P}}$ is a probability measure. Furthermore, if X is a nonnegative random variable, then

$$\tilde{\mathbb{E}}[X] = \mathbb{E}[XZ].$$

If Z is almost surely strictly positive, we also have

$$\mathbb{E}[Y] = \tilde{\mathbb{E}}\left[\frac{Y}{Z}\right]$$

for every nonnegative random variable Y .

Definition 1.6.3. Let Ω be a nonempty set and $\mathcal{F}$ a σ -algebra of subsets of Ω . Two probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F})$ are said to be *equivalent* if they agree on which sets in $\mathcal{F}$ have probability zero.

Definition 1.6.5. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\tilde{\mathbb{P}}$ be another probability measure on $(\Omega, \mathcal{F})$ that is equivalent to $\mathbb{P}$, and let Z be an almost surely positive random variable that relates $\mathbb{P}$ and $\tilde{\mathbb{P}}$ via Theorem 1.6.1. Then Z is called the *Radon-Nikodym derivative* of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$, and we write

$$Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}.$$

Theorem 1.6.7 (Radon-Nikodym). Let $\mathbb{P}$ and $\tilde{\mathbb{P}}$ be equivalent probability measures defined on $(\Omega, \mathcal{F})$. Then there exists an almost surely positive random variable Z such that $\mathbb{E}[Z] = 1$ and

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega) \quad \forall A \in \mathcal{F}.$$

1.2 Side Notes on Material

Measure Theory Books/Links

Terry Tao is among the most prominent mathematicians today, and his explanations are clear and to the point. Even the most accomplished mathematicians acknowledge that measure theory is far from intuitive—it is a difficult subject that requires substantial practice and time. The primary concern is reconciling notation that differs from Shreve's. He also has articles on his blog.

Axioms of Probability – there exists a “sample space” Ω (or S) containing all possible outcomes and a probability measure $\mathbb{P}$ that assigns a number in $[0, 1]$ to subsets of Ω :

(i) $\mathbb{P}(\Omega) = 1$

(ii) $E_1, E_2, \dots$ are disjoint events $\Rightarrow \mathbb{P}\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(E_n)$

*The main difference in a measure-theoretic setting is assigning the union of subsets/events to some σ -field, as depicted in Definition 1.1.2. In a discrete setting, where everything is countable and well-behaved, there is no need for this concept of a σ -field; however, the extension to the reals and higher dimensions creates complications.

Analysis — Sets, σ -Algebras, Measures

Borel σ -algebra (non-intuitive): Let O be the set of all open **sets** and C be the set of

all open **intervals**. Then the Borel σ -algebra is the smallest σ -algebra containing O . This tells us that: (1) since the σ -algebra contains every open set, it also contains every open interval (C); (2) since every open set is a countable union of open intervals, the σ -field generated by C contains all open sets in $\mathbb{R}$, so this field contains the σ -field generated by O . Therefore, the Borel σ -algebra is equivalent to the σ -algebra generated by open intervals C . A proof of the fact about countable unions can be found at: https://proofwiki.org/wiki/Open_Sets_in_Real_Number_Line

To summarize: the Borel σ -algebra of a set, say $\mathbb{R}$, is the smallest σ -algebra containing all the open intervals in $\mathbb{R}$. Note that the power set is the largest σ -algebra containing all the open intervals of $\mathbb{R}$. Since every open set is a union of a sequence of open intervals, and since the Borel σ -algebra is itself a σ -algebra, it contains all open sets. Closed intervals are complements of open intervals and closed sets are complements of open sets; consequently, by definition, the Borel σ -algebra contains all closed sets and closed intervals as well. This video explains why it is important to consider Borel σ -algebras rather than simply taking a larger σ -algebra containing all open sets: <https://www.youtube.com/watch?v=z5m6HXXx0Wo>

Other Intuitive Videos

1) What is a σ -algebra?

<https://www.youtube.com/watch?v=xZ69KEg7ccU>

2) What is a measure?

<https://www.youtube.com/watch?v=707qPrNIt7w>

For the enthusiastic reader, pages 16–24 of <http://assets.press.princeton.edu/chapters/s8008.pdf> make the notions of measurability and Borel sets/ σ -algebras more rigorous.

“With probability one” (a.s.) — the tricky part is that this does not imply other events cannot happen. Good example: pick a real number randomly between 0 and 10. Using a uniform (Lebesgue) measure, the probability of getting exactly 5 is 0, and its complement (anything else in the interval) occurs with probability 1. This does not mean it is theoretically impossible to pick 5, however. The second Borel-Cantelli lemma gives a more structured argument of this “almost surely” idea.

1.3 Potpourri — Other Useful Probability Concepts

Wikipedia searches should suffice to understand these: convolutions, convergence theorems (MCT, LDCT, BCT (LDCT with Y as a constant)), MGFs, Laplace transforms with respect to random variables, characteristic functions, PGFs, indicator properties and utilities, inequalities (SDP, Chebyshev, Chernoff bounds, etc.), CLT and variations (Lindeberg-Feller, Berry-Esseen, etc.), SLLN and its variations, etc.

1.4 Chapter 1 Exercises

Exercise 1.1. Using the properties of Definition 1.1.2 for a probability measure $\mathbb{P}$, show the following.

- (i) If $A \in \mathcal{F}$ and $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
(ii) If $A \in \mathcal{F}$ and $\{A_n\}_{n=1}^\infty$ is a sequence of sets in $\mathcal{F}$ with $\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = 0$ and $A \subset A_n \forall n$, then $\mathbb{P}(A) = 0$.

Solution

(i) $\mathbb{P}(B) = \mathbb{P}((B \cap A^c) \cup A)$. Since $(B \cap A^c) \cap A = \emptyset$ and $\mathbb{P}$ is nonnegative, countable additivity gives $\mathbb{P}(B) = \mathbb{P}(B \cap A^c) + \mathbb{P}(A) \geq \mathbb{P}(A)$.

(ii) $A_n \downarrow A \Rightarrow A_n^c \uparrow A^c$. Define an increasing sequence of events $\{E_n, n \geq 1\}$ by

$$E_1 = A_1^c, \text{ and } E_n = A_n^c \cap A_{n-1}, \quad n > 1.$$

These events are mutually exclusive, so

$$\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty A_i^c \quad \text{and} \quad \bigcup_{i=1}^n E_i = \bigcup_{i=1}^n A_i^c.$$

Therefore, $\mathbb{P}\left(\bigcup_{i=1}^\infty A_i^c\right) = \sum_{i=1}^\infty \mathbb{P}(E_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(E_i) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i=1}^n A_i^c\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n^c)$.

Since $\bigcup_1^\infty A_n^c = (\bigcap_1^\infty A_n)^c$,

$$1 - \mathbb{P}\left(\bigcap_{i=1}^\infty A_n\right) = \lim_{n \rightarrow \infty} (1 - \mathbb{P}(A_n)) \Rightarrow \mathbb{P}\left(\bigcap_{i=1}^\infty A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

We have shown that if $A_n \downarrow A$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$. Since $A \subset A_n$ for all n , we have $A_n \downarrow A$ (or to a set containing A), so $\mathbb{P}(A) \leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n) = 0$ by part (i) and the hypothesis. Hence $\mathbb{P}(A) = 0$.

Exercise 1.2. The infinite coin-toss space Ω_∞ of Example 1.1.4 is *uncountably infinite*. To see this, suppose there were a sequential list of all elements:

$$\begin{aligned} \omega^{(1)} &= \omega_1^{(1)} \omega_2^{(1)} \omega_3^{(1)} \omega_4^{(1)} \cdots, \\ \omega^{(2)} &= \omega_1^{(2)} \omega_2^{(2)} \omega_3^{(2)} \omega_4^{(2)} \cdots, \\ \omega^{(3)} &= \omega_1^{(3)} \omega_2^{(3)} \omega_3^{(3)} \omega_4^{(3)} \cdots, \\ &\vdots \end{aligned}$$

A sequence not in this list is obtained by flipping each diagonal entry $\omega_n^{(n)}$. Now consider

$$A = \{\omega = \omega_1 \omega_2 \omega_3 \omega_4 \cdots; \omega_1 = \omega_2, \omega_3 = \omega_4, \dots\}.$$

- (i) Show that A is uncountably infinite.
(ii) Show that when $p \in (0, 1)$, we have $\mathbb{P}(A) = 0$.

Solution

(i) Define a bijection $\varphi : A \rightarrow \Omega_\infty$ by mapping each sequence in A to the subsequence of its odd-indexed terms: $\varphi(\omega_1 \omega_1 \omega_2 \omega_2 \cdots) = \omega_1 \omega_2 \cdots$. This map is well-defined since A is precisely the set of sequences where $\omega_{2k-1} = \omega_{2k}$ for all $k \geq 1$, and it is clearly a bijection onto Ω_∞ . Since Ω_∞ is uncountably infinite, A is uncountably infinite.

(ii) Define the decreasing sequence of events

$$A_n = \{\omega : \omega_1 = \omega_2, \omega_3 = \omega_4, \dots, \omega_{2n-1} = \omega_{2n}\}.$$

Each pair agrees with probability $p^2 + (1 - p)^2$, so

$$\mathbb{P}(A_n) = (p^2 + (1 - p)^2)^n.$$

Since $p \in (0, 1)$, we have $p^2 + (1 - p)^2 < 1$, and thus $\mathbb{P}(A_n) \rightarrow 0$. Since $A_n \downarrow A$, continuity of $\mathbb{P}$ from above gives $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n) = 0$.

Exercise 1.3. Consider the set function $\mathbb{P}$ defined for every subset of $[0, 1]$ by $\mathbb{P}(A) = 0$ if A is a finite set and $\mathbb{P}(A) = \infty$ if A is an infinite set. Show that $\mathbb{P}$ satisfies (1.1.3)–(1.1.5), but $\mathbb{P}$ does not have the countable additivity property (1.1.2).

Solution

Properties (1.1.3)–(1.1.5) are clearly satisfied: $\mathbb{P}(\emptyset) = 0$ since $\emptyset$ is finite; $\mathbb{P}(A) \geq 0$ for all A ; and finite additivity holds since a finite union of finite sets is finite (with measure 0) and any union involving an infinite set is infinite (with measure ∞). For countable additivity (1.1.2), let $A_n = \{1/n\}$ for $n = 1, 2, \dots$. Each singleton is finite, so $\mathbb{P}(A_n) = 0$ for all n and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = 0$. However, $\bigcup_{n=1}^{\infty} A_n = \{1, 1/2, 1/3, \dots\}$ is an infinite set, so $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) = \infty \neq 0$. This violates countable additivity.

Exercise 1.4. (i) Construct a standard normal random variable Z on the probability space $(\Omega_{\infty}, \mathcal{F}_{\infty}, \mathbb{P})$ of Example 1.1.4 under the assumption that the probability of heads is $p = \frac{1}{2}$.

(ii) Define a sequence of random variables $\{Z_n\}_{n=1}^{\infty}$ on Ω_{∞} such that

$$\lim_{n \rightarrow \infty} Z_n(\omega) = Z(\omega) \quad \forall \omega \in \Omega_{\infty}$$

and, for each n , Z_n depends only on the first n coin tosses.

Solution

(i) In Example 1.2.5, Shreve constructs a uniform $(0, 1)$ random variable X on this probability space. Since the inverse normal CDF $N^{-1} : (0, 1) \rightarrow \mathbb{R}$ is well-defined, we may set $Z = N^{-1}(X)$. Example 1.2.6 verifies that Z is indeed a standard normal random variable on this space.

(ii) Define

$$Y_k(\omega) = \begin{cases} 1 & \text{if } \omega_k = H, \\ 0 & \text{if } \omega_k = T, \end{cases}$$

and set $X_n = \sum_{k=1}^n \frac{Y_k}{2^k}$, which approximates the uniform random variable $X = \sum_{k=1}^{\infty} \frac{Y_k}{2^k}$ using only the first n tosses. Since $X_n \uparrow X$ monotonically and N^{-1} is continuous and strictly increasing, the composition $Z_n := N^{-1}(X_n)$ is well-defined for $X_n \in (0, 1)$ and converges pointwise to $Z = N^{-1}(X)$ almost surely by the Monotone Convergence Theorem applied to $X_n \uparrow X$.

Exercise 1.5. Let X be a nonnegative random variable with cumulative distribution function $F(x) = \mathbb{P}\{X \leq x\}$. Show that

$$\mathbb{E}[X] = \int_0^\infty \bar{F}(x) dx$$

by showing that

$$\int_\Omega \int_0^\infty \mathbb{I}_{[0, X(\omega))}(x) dx d\mathbb{P}(\omega)$$

is equal to both $\mathbb{E}[X]$ and $\int_0^\infty \bar{F}(x) dx$.

Solution

Since $X \geq 0$ and $\mathbb{I}_{[0, X(\omega))}(x) \geq 0$ for all $x \in \mathbb{R}^+$, Tonelli's theorem applies:

$$\int_\Omega \int_0^\infty \mathbb{I}_{[0, X(\omega))}(x) dx d\mathbb{P}(\omega) = \int_\Omega X(\omega) d\mathbb{P}(\omega) = \mathbb{E}[X].$$

Switching the order of integration (justified by Tonelli),

$$\int_\Omega \int_0^\infty \mathbb{I}_{[0, X(\omega))}(x) dx d\mathbb{P}(\omega) = \int_0^\infty \int_\Omega \mathbb{I}_{[0, X(\omega))}(x) d\mathbb{P}(\omega) dx = \int_0^\infty \mathbb{P}\{X > x\} dx = \int_0^\infty \bar{F}(x) dx,$$

where the second equality uses $\mathbb{I}_{[0, X(\omega))}(x) = 1 \iff x < X(\omega)$, i.e., $\iff X(\omega) > x$.

Exercise 1.6. Let u be a fixed number in $\mathbb{R}$, and define the convex function $\phi(x) = e^{ux}$ for all $x \in \mathbb{R}$. Let X be a normal random variable with mean $\mu := \mathbb{E}[X]$ and standard deviation $\sigma := [\mathbb{E}[(X - \mu)^2]]^{1/2}$, i.e., with density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}.$$

- (i) Verify that $\mathbb{E}[e^{uX}] = e^{u\mu + \frac{1}{2}u^2\sigma^2}$.
- (ii) Verify that Jensen's inequality holds: $\mathbb{E}[\phi(X)] \geq \phi(\mathbb{E}[X])$.

Solution

(i) The key technique is completing the square in the exponent.

$$\mathbb{E}[e^{uX}] = \frac{1}{\sigma\sqrt{2\pi}} \int_{\mathbb{R}} e^{ux} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx.$$

Combining the exponents:

$$ux - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2}(x^2 - 2\mu x + \mu^2 - 2\sigma^2 ux) = -\frac{1}{2\sigma^2}(x - (\mu + \sigma^2 u))^2 + u\mu + \frac{1}{2}\sigma^2 u^2.$$

Substituting $z = (x - (\mu + \sigma^2 u))/\sigma$,

$$\mathbb{E}[e^{uX}] = \frac{1}{\sqrt{2\pi}} e^{u\mu + \frac{1}{2}\sigma^2 u^2} \int_{\mathbb{R}} e^{-z^2/2} dz = e^{u\mu + \frac{1}{2}\sigma^2 u^2}.$$

(ii) $\mathbb{E}[\phi(X)] = e^{u\mu + \frac{1}{2}\sigma^2 u^2}$ and $\phi(\mathbb{E}[X]) = \phi(\mu) = e^{u\mu}$. Since $\frac{1}{2}\sigma^2 u^2 \geq 0$ for all $\sigma, u \in \mathbb{R}$, we have $e^{u\mu + \frac{1}{2}\sigma^2 u^2} \geq e^{u\mu}$, confirming Jensen's inequality.

Exercise 1.7. For each positive integer n , define f_n to be the normal density with mean zero and variance n :

$$f_n(x) = \frac{1}{\sqrt{2n\pi}} e^{-x^2/(2n)}.$$

- (i) What is the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$?
- (ii) What is $\lim_{n \rightarrow \infty} \int_{-\infty}^\infty f_n(x) dx$?
- (iii) Explain why $\lim_{n \rightarrow \infty} \int_{-\infty}^\infty f_n(x) dx \neq \int_{-\infty}^\infty f(x) dx$ does not violate the Monotone Con-

vergence Theorem or the Dominated Convergence Theorem.

Solution

(i) $f(x) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2n\pi}} e^{-x^2/(2n)} = 0$ for every $x \in \mathbb{R}$, since the prefactor $1/\sqrt{2n\pi} \rightarrow 0$ and the exponential factor tends to 1.

(ii) Setting $z = x/\sqrt{n}$ so that $dx = \sqrt{n} dz$,

$$\int_{-\infty}^{\infty} f_n(x) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz = 1,$$

so the limit is 1 for every n and hence $\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n(x) dx = 1$.

(iii) The Monotone Convergence Theorem is not violated because the sequence $\{f_n\}$ is not monotone: $f_n(x)$ increases near the origin and decreases in the tails as n grows, so the pointwise ordering $f_1 \leq f_2 \leq \dots$ does not hold. The Dominated Convergence Theorem is not violated because there is no integrable dominating function g satisfying $|f_n(x)| \leq g(x)$ a.e. for all n : the sequence “spreads out” its unit mass, so any such g would have to have infinite integral.

Exercise 1.8 (Moment-Generating Function).

(i) Use the Dominated Convergence Theorem (Theorem 1.4.9) and equation (1.9.1) to show that

$$\lim_{n \rightarrow \infty} \mathbb{E}[Y_n] = \mathbb{E}\left[\lim_{n \rightarrow \infty} Y_n\right] = \mathbb{E}[Xe^{tX}].$$

This establishes the desired formula $\phi'(t) = \mathbb{E}[Xe^{tX}]$.

(ii) Suppose the random variable X can take both positive and negative values and $\mathbb{E}[e^{tX}] < \infty$ and $\mathbb{E}[|X|e^{tX}] < \infty$ for all $t \in \mathbb{R}$. Show that $\phi'(t) = \mathbb{E}[Xe^{tX}]$.

Solution

(i) By the difference quotient definition, $Y_n = \frac{e^{tX} - e^{s_n X}}{t - s_n}$ where $s_n \rightarrow t$. By the Mean Value Theorem, $Y_n = Xe^{\theta_n X}$ for some θ_n between s_n and t . For $X \geq 0$, this gives $|Y_n| = |X|e^{\theta_n X} \leq |X|e^{\alpha X}$ for any fixed $\alpha \geq t$, and $\mathbb{E}[|X|e^{\alpha X}] < \infty$ by hypothesis. For $X < 0$, a symmetric bound applies using the lower endpoint. The dominating function is integrable, so DCT applies and yields $\phi'(t) = \mathbb{E}[Xe^{tX}]$.

(ii) Write $X = X^+ - X^-$ where $X^+ = \max\{X, 0\}$ and $X^- = \max\{-X, 0\}$ are nonnegative. The difference quotient $Y_n = (e^{tX} - e^{s_n X})/(t - s_n)$ splits via linearity, and on each of $\{X \geq 0\}$ and $\{X < 0\}$ the same MVT bound applies: $|Y_n| \leq |X|e^{\alpha|X|}$ for $\alpha = \max(|t|, |s_n|)$. By hypothesis $\mathbb{E}[|X|e^{tX}] < \infty$ controls both parts, so DCT again gives $\phi'(t) = \mathbb{E}[Xe^{tX}]$.

Exercise 1.9. Suppose X is a random variable on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$, A is a set in $\mathcal{F}$, and for every Borel subset $\mathcal{B}$ of $\mathbb{R}$,

$$\int_A \mathbb{I}_{\mathcal{B}}(X(\omega)) d\mathbb{P}(\omega) = \mathbb{P}(A) \cdot \mathbb{P}\{X \in \mathcal{B}\}.$$

We say X is *independent* of the event A . Show that if X is independent of A , then

$$\int_A g(X(\omega)) d\mathbb{P}(\omega) = \mathbb{P}(A) \cdot \mathbb{E}[g(X)]$$

for every nonnegative, Borel-measurable function g .

Solution

We prove this using the standard machine.

Step 1: Indicator functions. If $g(x) = \mathbb{I}_B(x)$ for some Borel set B , the result is exactly the given independence condition:

$$\int_A \mathbb{I}_B(X(\omega)) d\mathbb{P}(\omega) = \mathbb{P}(A) \cdot \mathbb{P}\{X \in B\} = \mathbb{P}(A) \cdot \mathbb{E}[\mathbb{I}_B(X)].$$

Step 2: Nonnegative simple functions. If $g(x) = \sum_{k=1}^n \alpha_k \mathbb{I}_{B_k}(x)$ with $\alpha_k \geq 0$, linearity of the integral and Step 1 give:

$$\int_A g(X(\omega)) d\mathbb{P}(\omega) = \sum_{k=1}^n \alpha_k \int_A \mathbb{I}_{B_k}(X(\omega)) d\mathbb{P}(\omega) = \sum_{k=1}^n \alpha_k \mathbb{P}(A) \cdot \mathbb{E}[\mathbb{I}_{B_k}(X)] = \mathbb{P}(A) \cdot \mathbb{E}[g(X)].$$

Step 3: Nonnegative Borel-measurable functions. Any such g is the pointwise monotone limit of nonnegative simple functions $g_n \uparrow g$. By Step 2 and Monotone Convergence (applied separately to each side),

$$\int_A g(X(\omega)) d\mathbb{P}(\omega) = \lim_{n \rightarrow \infty} \int_A g_n(X(\omega)) d\mathbb{P}(\omega) = \lim_{n \rightarrow \infty} \mathbb{P}(A) \cdot \mathbb{E}[g_n(X)] = \mathbb{P}(A) \cdot \mathbb{E}[g(X)].$$

Exercise 1.10. Let $\mathbb{P}$ be the uniform (Lebesgue) measure on $\Omega = [0, 1]$. Define

$$Z(\omega) = \begin{cases} 0 & \text{if } \omega \in [0, \frac{1}{2}), \\ 2 & \text{if } \omega \in [\frac{1}{2}, 1]. \end{cases}$$

For $A \in \mathcal{B}[0, 1]$, define $\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega)$.

- (i) Show that $\tilde{\mathbb{P}}$ is a probability measure.
 - (ii) Show that $\mathbb{P}(A) = 0$ implies $\tilde{\mathbb{P}}(A) = 0$.
 - (iii) Show that there is a set A for which $\tilde{\mathbb{P}}(A) = 0$ but $\mathbb{P}(A) > 0$.
-

Solution

(i) $\tilde{\mathbb{P}}(\Omega) = \mathbb{E}[Z] = 2 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = 1$. Countable additivity: for disjoint sets $A_1, A_2, \dots$, let $B_n = \bigcup_{k=1}^n A_k$. Since $Z \geq 0$, $\mathbb{I}_{B_n} Z \uparrow \mathbb{I}_{B_\infty} Z$ monotonically, so by MCT:

$$\tilde{\mathbb{P}}(\bigcup_{k=1}^\infty A_k) = \int_\Omega \mathbb{I}_{B_\infty} Z d\mathbb{P} = \lim_{n \rightarrow \infty} \int_\Omega \sum_{k=1}^n \mathbb{I}_{A_k} Z d\mathbb{P} = \sum_{k=1}^\infty \tilde{\mathbb{P}}(A_k).$$

(ii) If $\mathbb{P}(A) = 0$, then $\mathbb{P}(A \cap [\frac{1}{2}, 1]) \leq \mathbb{P}(A) = 0$, so $\tilde{\mathbb{P}}(A) = 2 \mathbb{P}(A \cap [\frac{1}{2}, 1]) = 0$.

(iii) Let $A = [0, \frac{1}{2})$. Then $Z(\omega) = 0$ for all $\omega \in A$, so $\tilde{\mathbb{P}}(A) = 0$, but $\mathbb{P}(A) = \frac{1}{2} > 0$.

Exercise 1.11. In Example 1.6.6, X is standard normal under $\mathbb{P}$, $Y = X + \theta$ for a constant θ , $Z = e^{-\theta X - \frac{1}{2}\theta^2}$, and $\tilde{\mathbb{P}}(A) = \int_A Z d\mathbb{P}$. Give another proof that Y is standard normal under $\tilde{\mathbb{P}}$ by verifying

$$\tilde{\mathbb{E}}[e^{uY}] = e^{\frac{1}{2}u^2} \text{ for all } u \in \mathbb{R}.$$

Solution

Using Theorem 1.6.1 with $\tilde{\mathbb{E}}[e^{uY}] = \mathbb{E}[Ze^{uY}]$:

$$\tilde{\mathbb{E}}[e^{uY}] = \mathbb{E}\left[e^{-\theta X - \frac{1}{2}\theta^2} \cdot e^{u(X+\theta)}\right] = \mathbb{E}\left[e^{(u-\theta)X}\right] \cdot e^{u\theta - \frac{1}{2}\theta^2}.$$

Since X is standard normal under $\mathbb{P}$, $\mathbb{E}[e^{(u-\theta)X}] = e^{\frac{1}{2}(u-\theta)^2}$ by Exercise 1.6. Therefore:

$$\tilde{\mathbb{E}}[e^{uY}] = e^{\frac{1}{2}(u-\theta)^2} \cdot e^{u\theta - \frac{1}{2}\theta^2} = \exp\left\{\frac{1}{2}u^2 - u\theta + \frac{1}{2}\theta^2 + u\theta - \frac{1}{2}\theta^2\right\} = e^{\frac{1}{2}u^2}.$$

Since $e^{u^2/2}$ is the moment-generating function of a standard normal, Y is standard normal under $\tilde{\mathbb{P}}$.

Exercise 1.12. In the setup of Exercise 1.11, now define $\hat{Z} = e^{\theta Y - \frac{1}{2}\theta^2}$ and

$$\hat{\mathbb{P}}(A) = \int_A \hat{Z}(\omega) d\tilde{\mathbb{P}}(\omega) \text{ for all } A \in \mathcal{F},$$

so that $X = Y - \theta$ is standard normal under $\hat{\mathbb{P}}$. Show that $\hat{Z} = 1/Z$ and $\hat{\mathbb{P}} = \mathbb{P}$.

Solution

Since $Y = X + \theta$,

$$\hat{Z} = e^{\theta Y - \frac{1}{2}\theta^2} = e^{\theta(X+\theta) - \frac{1}{2}\theta^2} = e^{\theta X + \frac{1}{2}\theta^2} = \frac{1}{e^{-\theta X - \frac{1}{2}\theta^2}} = \frac{1}{Z}.$$

For any $A \in \mathcal{F}$, using Theorem 1.6.1:

$$\hat{\mathbb{P}}(A) = \int_A \hat{Z} d\tilde{\mathbb{P}} = \tilde{\mathbb{E}}[\mathbb{I}_A \cdot \frac{1}{Z}] = \mathbb{E}[Z \cdot \mathbb{I}_A \cdot \frac{1}{Z}] = \mathbb{E}[\mathbb{I}_A] = \mathbb{P}(A).$$

Exercise 1.13 (Change of Measure for a Normal Random Variable). With X standard normal under $\mathbb{P}$, $Y = X + \theta$, and $A(\bar{\omega}, \epsilon) = \{X \in B(x, \epsilon)\} = \{Y \in B(y, \epsilon)\}$ where $x = X(\bar{\omega})$, $y = x + \theta$:

- (i) Show that $\frac{1}{\epsilon} \mathbb{P}\{X \in B(x, \epsilon)\} \approx \frac{1}{\sqrt{2\pi}} \exp\{-X^2(\bar{\omega})/2\}$.
 - (ii) Show that we must have $\frac{1}{\epsilon} \tilde{\mathbb{P}}\{Y \in B(y, \epsilon)\} \approx \frac{1}{\sqrt{2\pi}} \exp\{-Y^2(\bar{\omega})/2\}$ for Y to be standard normal under $\tilde{\mathbb{P}}$.
 - (iii) Show that $\{X \in B(x, \epsilon)\}$ and $\{Y \in B(y, \epsilon)\}$ are the same set.
 - (iv) Show that $\frac{\tilde{\mathbb{P}}(A(\bar{\omega}, \epsilon))}{\mathbb{P}(A(\bar{\omega}, \epsilon))} \approx \exp\{-\theta X(\bar{\omega}) - \frac{1}{2}\theta^2\}$.
-

Solution

- (i) By the Mean Value Theorem applied to $\int_{x-\epsilon/2}^{x+\epsilon/2} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$:

$$\frac{1}{\epsilon} \mathbb{P}\{X \in B(x, \epsilon)\} = \frac{1}{\epsilon} \int_{x-\epsilon/2}^{x+\epsilon/2} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \approx \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{X^2(\bar{\omega})}{2}\right\}.$$

- (ii) For Y to be standard normal under $\tilde{\mathbb{P}}$, its density must be $(1/\sqrt{2\pi})e^{-y^2/2}$, so by the same argument:

$$\frac{1}{\epsilon} \tilde{\mathbb{P}}\{Y \in B(y, \epsilon)\} \approx \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{Y^2(\bar{\omega})}{2}\right\}.$$

- (iii) $\{Y \in B(y, \epsilon)\} = \{X + \theta \in B(x + \theta, \epsilon)\} = \{X \in B(x, \epsilon)\}$, since adding θ to X shifts the interval by θ .

- (iv) Dividing the approximations from (ii) and (i):

$$\frac{\tilde{\mathbb{P}}(A(\bar{\omega}, \epsilon))}{\mathbb{P}(A(\bar{\omega}, \epsilon))} \approx \frac{e^{-Y^2(\bar{\omega})/2}}{e^{-X^2(\bar{\omega})/2}} = e^{(X^2(\bar{\omega}) - Y^2(\bar{\omega}))/2}.$$

Since $Y(\bar{\omega}) = X(\bar{\omega}) + \theta$,

$$X^2 - Y^2 = X^2 - (X + \theta)^2 = -2\theta X - \theta^2,$$

so $\frac{\tilde{\mathbb{P}}(A)}{\mathbb{P}(A)} \approx e^{-\theta X(\bar{\omega}) - \theta^2/2}$, which is exactly $Z(\bar{\omega})$ from Example 1.6.6.

Exercise 1.14 (Change of Measure for an Exponential Random Variable). Let X have the exponential distribution $\mathbb{P}\{X \leq a\} = 1 - e^{-\lambda a}$ for $a \geq 0$, where $\lambda > 0$. Let $\tilde{\lambda} > 0$ and define $Z = \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)X}$ and $\tilde{\mathbb{P}}(A) = \int_A Z d\mathbb{P}$.

- (i) Show that $\tilde{\mathbb{P}}(\Omega) = 1$.
(ii) Compute the CDF of X under $\tilde{\mathbb{P}}$.
-

Solution

- (i) Since X has density $f(x) = \lambda e^{-\lambda x}$ on $x \geq 0$:

$$\tilde{\mathbb{P}}(\Omega) = \mathbb{E}[Z] = \int_0^\infty \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)x} \cdot \lambda e^{-\lambda x} dx = \tilde{\lambda} \int_0^\infty e^{-\tilde{\lambda}x} dx = \tilde{\lambda} \cdot \frac{1}{\tilde{\lambda}} = 1.$$

- (ii) For $a \geq 0$:

$$\tilde{\mathbb{P}}\{X \leq a\} = \int_0^a \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda}-\lambda)x} \cdot \lambda e^{-\lambda x} dx = \tilde{\lambda} \int_0^a e^{-\tilde{\lambda}x} dx = 1 - e^{-\tilde{\lambda}a}.$$

Thus under $\tilde{\mathbb{P}}$, X has the exponential distribution with parameter $\tilde{\lambda}$.

Exercise 1.15 (Provided by Alexander Ng). Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ with strictly positive density $f(x)$. Let g be strictly increasing, differentiable, with $g(y) \rightarrow \pm\infty$ as $y \rightarrow \pm\infty$, and define $Y = g(X)$. Let $h(y) \geq 0$ with $\int_{-\infty}^\infty h(y) dy = 1$. Define

$$Z = \frac{h(g(X))g'(X)}{f(X)}.$$

- (i) Show that $Z \geq 0$ and $\mathbb{E}[Z] = 1$.
(ii) Define $\tilde{\mathbb{P}}(A) = \int_A Z d\mathbb{P}$ for all $A \in \mathcal{F}$. Show that Y has density h under $\tilde{\mathbb{P}}$.
-

Solution

- (i) Since g is strictly increasing, $g'(X) \geq 0$; since $h \geq 0$ and $f > 0$, we have $Z \geq 0$. Using the change-of-variables formula with $y = g(x)$, $dy = g'(x) dx$:

$$\mathbb{E}[Z] = \int_{-\infty}^\infty \frac{h(g(x))g'(x)}{f(x)} \cdot f(x) dx = \int_{-\infty}^\infty h(g(x))g'(x) dx = \int_{-\infty}^\infty h(y) dy = 1.$$

- (ii) For any Borel set $B \subset \mathbb{R}$,

$$\begin{aligned} \tilde{\mathbb{P}}\{Y \in B\} &= \mathbb{E}[Z \cdot \mathbb{I}_B(Y)] = \int_{-\infty}^\infty \mathbb{I}_B(g(x)) \frac{h(g(x))g'(x)}{f(x)} \cdot f(x) dx \\ &= \int_{-\infty}^\infty \mathbb{I}_B(g(x)) h(g(x)) g'(x) dx = \int_B h(y) dy, \end{aligned}$$

where the last step uses the substitution $y = g(x)$ and $g' > 0$. Since $\tilde{\mathbb{P}}\{Y \in B\} = \int_B h(y) dy$ for all Borel B , the density of Y under $\tilde{\mathbb{P}}$ is h .

2. Information and Conditioning

2.1 Definitions, Theorems, Corollaries

Definition 2.1.1. Let Ω be a nonempty set. Let T be a fixed positive number, and assume that for each $t \in [0, T]$ there is a σ -algebra $\mathcal{F}(t)$. Assume further that if $s \leq t$, then every set in $\mathcal{F}(s)$ is also in $\mathcal{F}(t)$. Then we call the collection of σ -algebras $\mathcal{F}(t)$, $0 \leq t \leq T$, a *filtration*.

Definition 2.1.3. Let X be a random variable defined on a nonempty sample space Ω . The σ -algebra generated by X , denoted $\sigma(X)$, is the collection of all subsets of Ω of the form $\{X \in B\}$, where B ranges over the Borel subsets of $\mathbb{R}$.

Definition 2.1.5. Let X be a random variable defined on a nonempty sample space Ω . Let $\mathcal{G}$ be a σ -algebra of subsets of Ω . If every set in $\sigma(X)$ is also in $\mathcal{G}$, we say that X is $\mathcal{G}$ -measurable.

Definition 2.1.6. Let Ω be a nonempty sample space equipped with a filtration $\mathcal{F}(t)$, $0 \leq t \leq T$. Let $X(t)$ be a collection of random variables indexed by $t \in [0, T]$. We say this collection of random variables is an *adapted stochastic process* if, for each t , the random variable $X(t)$ is $\mathcal{F}(t)$ -measurable.

Definition 2.2.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{G}$ and $\mathcal{H}$ be sub- σ -algebras of $\mathcal{F}$. We say these two σ -algebras are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \quad \forall A \in \mathcal{G}, B \in \mathcal{H}.$$

Let X and Y be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. We say X and Y are *independent* if $\sigma(X)$ and $\sigma(Y)$ are independent. We say that X is independent of the σ -algebra $\mathcal{G}$ if $\sigma(X)$ and $\mathcal{G}$ are independent.

Definition 2.2.3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3, \dots$ be a sequence of sub- σ -algebras of $\mathcal{F}$. We say that the n σ -algebras $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent if

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2) \cdot \dots \cdot \mathbb{P}(A_n)$$

for all $A_1 \in \mathcal{G}_1, \dots, A_n \in \mathcal{G}_n$. The full sequence $\mathcal{G}_1, \mathcal{G}_2, \dots$ is independent if, for every positive integer n , the n σ -algebras $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent. Independence of a sequence of random variables $X_1, X_2, \dots$ is defined analogously via $\sigma(X_1), \sigma(X_2), \dots$.

Theorem 2.2.5. Let X and Y be independent random variables, and let f and g be Borel-measurable functions on $\mathbb{R}$. Then $f(X)$ and $g(Y)$ are independent random variables.

Definition 2.2.6. Let X and Y be random variables. The pair (X, Y) takes values in $\mathbb{R}^2$, and the *joint distribution measure* of (X, Y) is

$$\mu_{X,Y}(C) = \mathbb{P}\{(X, Y) \in C\} \text{ for all Borel sets } C \subset \mathbb{R}^2.$$

The *joint CDF* is $F_{X,Y}(a, b) = \mathbb{P}\{X \leq a, Y \leq b\}$. A nonnegative Borel-measurable func-

tion $f_{X,Y}(x, y)$ is a *joint density* for (X, Y) if

$$F_{X,Y}(a, b) = \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx \text{ for all } a, b \in \mathbb{R}.$$

The *marginal* distributions, CDFs, and densities (when they exist) are obtained by integrating out the other variable.

Theorem 2.2.7. Let X and Y be random variables. The following conditions are equivalent.

- (i) X and Y are independent.
- (ii) The joint distribution measure factors: $\mu_{X,Y}(A \times B) = \mu_X(A) \cdot \mu_Y(B)$ for all Borel sets $A, B \subset \mathbb{R}$.
- (iii) The joint CDF factors: $F_{X,Y}(a, b) = F_X(a) \cdot F_Y(b)$ for all $a, b \in \mathbb{R}$.
- (iv) The joint MGF factors: $\mathbb{E}[e^{uX+vY}] = \mathbb{E}[e^{uX}] \cdot \mathbb{E}[e^{vY}]$ for all $u, v \in \mathbb{R}$ for which expectations are finite.

If there is a joint density, each of the above is also equivalent to:

- (v) The joint density factors: $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$ for almost every (x, y) .

The conditions above imply but are not equivalent to:

- (vi) $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$, provided $\mathbb{E}[|XY|] < \infty$.

Definition 2.2.11. Two random variables X and Y are said to be *jointly normal* if they have the joint density

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right]\right\},$$

where $\sigma_1 > 0$, $\sigma_2 > 0$, $|\rho| < 1$, and $\mu_1, \mu_2 \in \mathbb{R}$. More generally, a random column vector $\mathbf{X} = (X_1, \dots, X_n)^\top$ is *jointly normal* if it has joint density

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det(C)}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \mu)^\top C^{-1}(\mathbf{x} - \mu)\right\},$$

where C is the covariance matrix and μ is the mean vector.

Definition 2.3.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\mathcal{G}$ be a sub- σ -algebra of $\mathcal{F}$, and let X be a random variable that is either nonnegative or integrable. The *conditional expectation* of X given $\mathcal{G}$, denoted $\mathbb{E}[X|\mathcal{G}]$, is any random variable satisfying:

- (i) **(Measurability)** $\mathbb{E}[X|\mathcal{G}]$ is $\mathcal{G}$ -measurable, and
- (ii) **(Partial averaging)**

$$\int_A \mathbb{E}[X|\mathcal{G}](\omega) d\mathbb{P}(\omega) = \int_A X(\omega) d\mathbb{P}(\omega) \text{ for all } A \in \mathcal{G}.$$

Theorem 2.3.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G}$ a sub- σ -algebra of $\mathcal{F}$.

- (i) **(Linearity)** $\mathbb{E}[c_1X + c_2Y|\mathcal{G}] = c_1\mathbb{E}[X|\mathcal{G}] + c_2\mathbb{E}[Y|\mathcal{G}]$.
- (ii) **(Taking out what is known)** If X is $\mathcal{G}$ -measurable and Y , XY are integrable, then $\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$.
- (iii) **(Iterated conditioning)** If $\mathcal{H} \subset \mathcal{G}$ and X is integrable, then $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$.
- (iv) **(Independence)** If X is integrable and independent of $\mathcal{G}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.
- (v) **(Conditional Jensen's inequality)** If ϕ is convex and X is integrable, then $\mathbb{E}[\phi(X)|\mathcal{G}] \geq \phi(\mathbb{E}[X|\mathcal{G}])$.

Lemma 2.3.4 (Independence). Let $X_1, \dots, X_K$ be $\mathcal{G}$ -measurable and $Y_1, \dots, Y_L$ independent of $\mathcal{G}$. Define $g(x_1, \dots, x_K) = \mathbb{E}[f(x_1, \dots, x_K, Y_1, \dots, Y_L)]$. Then

$$\mathbb{E}[f(X_1, \dots, X_K, Y_1, \dots, Y_L) | \mathcal{G}] = g(X_1, \dots, X_K).$$

Definition 2.3.5. Let $M(t)$, $0 \leq t \leq T$, be an adapted stochastic process.

- (i) If $\mathbb{E}[M(t) | \mathcal{F}(s)] = M(s)$ for all $0 \leq s \leq t \leq T$, then M is a *martingale* (no tendency to rise or fall).
- (ii) If $\mathbb{E}[M(t) | \mathcal{F}(s)] \geq M(s)$, then M is a *submartingale* (no tendency to fall).
- (iii) If $\mathbb{E}[M(t) | \mathcal{F}(s)] \leq M(s)$, then M is a *supermartingale* (no tendency to rise).

Definition 2.3.6. An adapted process $X(t)$, $0 \leq t \leq T$, is a *Markov process* if for all $0 \leq s \leq t \leq T$ and every nonnegative Borel-measurable function f , there exists a Borel-measurable function g such that

$$\mathbb{E}[f(X(t)) | \mathcal{F}(s)] = g(X(s)).$$

2.2 Side Notes on Material

Markov Processes. Markov processes are the continuous-time analogs of Markov chains (CTMCs). Before handling CTMCs, it is useful to understand discrete-time Markov chains (DTMCs) to build intuition.

A DTMC is a stochastic process $\{X_n, n \in \mathbb{Z}^+\}$ taking values in a countable state space, possessing the Markov property:

$$\mathbb{P}\{X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0\} = \mathbb{P}\{X_{n+1} = j | X_n = i\} = p_{ij}.$$

Example: Gambler's Ruin Problem.

A gambler starting with $\$x$ wants to reach $\$N$. On each play, he wins $\$1$ with probability p or loses $\$1$ with probability $1 - p$. The game ends when his wealth is 0 or N . The probability of reaching $\$N$ before 0, starting from $\$x$, is $\frac{1-(q/p)^x}{1-(q/p)^N}$ when $p \neq q$ and x/N when $p = q = 1/2$, where $q = 1 - p$.

Chapman-Kolmogorov Equations and Classification of States.

Key definitions and notation:

- State j is *accessible* from state i if $\exists n \geq 0 : p_{ij}^{(n)} > 0$.
- States i and j *communicate* ($i \leftrightarrow j$) if each is accessible from the other.
- If there is a unique equivalence class, the chain is *irreducible*.
- The *period* of state i is $d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$; state i is *aperiodic* if $d(i) = 1$.
- $f_{ij}^{(n)} := \mathbb{P}(X_n = j, X_k \neq j \ \forall k \in \{1, \dots, n-1\} | X_0 = i)$ is the first-passage probability.
- State i is *recurrent* if $\sum_n f_{ii}^{(n)} = 1$, and *transient* otherwise.
- *Polya's Theorem*: simple random walks on $\mathbb{Z}^d$ are recurrent if and only if $d \leq 2$.
- A recurrent state i is *positive recurrent* if the expected return time $\mu_{ii} < \infty$, and *null recurrent* otherwise.
- A distribution $\{\pi_j\}$ is *stationary* if $\pi_j = \sum_i \pi_i p_{ij}$ for all j .
- For irreducible, aperiodic chains: $\pi_j = \lim_{n \rightarrow \infty} p_{ij}^{(n)} = 1/\mu_{jj}$.

CTMCs.

A CTMC is a stochastic process such that each time it enters state i : (i) the sojourn time is exponentially distributed with rate v_i ; (ii) upon leaving, the next state j is chosen with probability p_{ij} (a DTMC transition). A comprehensive reference on CTMCs:

<https://cs.nyu.edu/mishra/COURSES/09.HPGP/scribe3>

Filtration.

The filtration $(\mathcal{F}_t)_{t \geq 0}$ is a nondecreasing family of sub- σ -algebras encoding the information available up to time t . An active research area concerns filtration behavior under changes of measure and in insider trading models:

Survey: <https://arxiv.org/pdf/0712.0622.pdf>

Insider Trading Model: http://stats.lse.ac.uk/cetin/files/campi_cetin.pdf

Iterated and Conditional Expectation.

Simple, intuitive example: Suppose I roll a fair die and you must estimate the outcome X to minimize mean squared error. Your best guess is $\mathbb{E}[X] = 3.5$.

Now suppose I tell you X is even. The updated estimate becomes $(2 + 4 + 6)/3 = 4$; if X is odd, the estimate is $(1 + 3 + 5)/3 = 3$.

More generally, when the parity is revealed (itself random), your response is a function of that information—reflected in the requirement that $\mathbb{E}[X|\mathcal{G}]$ be $\mathcal{G}$ -measurable.

For further development, see pages 66–68 of Shreve, or this video (40:30–48:30):

https://www.youtube.com/watch?v=KEY8P4Rz4Ao&list=PLbMVogVj5nJQqGHrpAloTec_10KsG-foc&index=37

An example of partial averaging and $\mathcal{G}$ -measurable conditional expectation (page 116):

https://web.ma.utexas.edu/users/gordanz/notes/theory_of_probability_I.pdf

Martingales.

Martingales model fair games: the expected value of the process at any future time, given current information, equals the current value. Submartingales model “favorable” games (no tendency to fall); supermartingales model unfavorable ones (no tendency to rise).

Stopping times. A random variable $\tau \geq 0$ is a stopping time with respect to $\{\mathcal{F}_t\}$ if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

Optional Stopping Theorem (discrete time). Let $\{M(t)\}$ be a martingale with stopping time τ , and define the stopped process $\bar{M}(t) = M(\min\{t, \tau\})$. If any of the following hold: (i) $\bar{M}(t)$ is uniformly bounded; (ii) τ is bounded; or (iii) $\mathbb{E}[\tau] < \infty$ and increments are bounded in conditional expectation; then $\mathbb{E}[M(\tau)] = \mathbb{E}[M(0)]$.

Further reading: <http://math.bu.edu/people/prakashb/Math/continuous-time%20martingales.pdf>

2.3 Chapter 2 Exercises

Exercise 2.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a general probability space, and suppose a random variable X on this space is measurable with respect to the trivial σ -algebra $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Show that

X is not random (i.e., there is a constant c such that $X(\omega) = c$ for all $\omega \in \Omega$). Such a random variable is called degenerate.

Solution

Suppose for contradiction that $\exists \omega_1, \omega_2 \in \Omega$ with $X(\omega_1) \neq X(\omega_2)$. Setting $c_0 = X(\omega_1)$, the Borel set $B = \{c_0\}$ gives $\{X \in B\} = X^{-1}(B)$, which contains ω_1 but not ω_2 . Hence $X^{-1}(B) \notin \{\emptyset, \Omega\} = \mathcal{F}_0$, contradicting the assumption that X is $\mathcal{F}_0$ -measurable. Therefore $X(\omega) = c$ for all $\omega \in \Omega$.

Exercise 2.2. Consider the two-coin-toss space $\Omega_2 = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}$ with stock prices

$$\begin{aligned} S_0 &= 4, \quad S_1(\text{H}) = 8, \quad S_1(\text{T}) = 2, \\ S_2(\text{HH}) &= 16, \quad S_2(\text{HT}) = S_2(\text{TH}) = 4, \quad S_2(\text{TT}) = 1. \end{aligned}$$

Consider two probability measures:

$$\begin{aligned} \tilde{\mathbb{P}}(\text{HH}) &= \tilde{\mathbb{P}}(\text{HT}) = \tilde{\mathbb{P}}(\text{TH}) = \tilde{\mathbb{P}}(\text{TT}) = \frac{1}{4}, \\ \mathbb{P}(\text{HH}) &= \frac{4}{9}, \quad \mathbb{P}(\text{HT}) = \frac{2}{9}, \quad \mathbb{P}(\text{TH}) = \frac{2}{9}, \quad \mathbb{P}(\text{TT}) = \frac{1}{9}. \end{aligned}$$

Define

$$X = \begin{cases} 1 & \text{if } S_2 = 4, \\ 0 & \text{if } S_2 \neq 4. \end{cases}$$

- (i) List all the sets in $\sigma(X)$.
 - (ii) List all the sets in $\sigma(S_1)$.
 - (iii) Show that $\sigma(X)$ and $\sigma(S_1)$ are independent under $\tilde{\mathbb{P}}$.
 - (iv) Show that $\sigma(X)$ and $\sigma(S_1)$ are not independent under $\mathbb{P}$.
 - (v) Under $\mathbb{P}$, explain intuitively why knowing $X = 1$ should cause you to revise your estimate of the distribution of S_1 .
-

Solution

- (i) $X = 1$ for $\{\text{HT}, \text{TH}\}$ and $X = 0$ for $\{\text{HH}, \text{TT}\}$, so $\sigma(X) = \{\emptyset, \{\text{HT}, \text{TH}\}, \{\text{HH}, \text{TT}\}, \Omega\}$.
- (ii) $S_1 = 8$ on $\{\text{HH}, \text{HT}\}$ and $S_1 = 2$ on $\{\text{TH}, \text{TT}\}$, so $\sigma(S_1) = \{\emptyset, \{\text{HH}, \text{HT}\}, \{\text{TH}, \text{TT}\}, \Omega\}$.
- (iii) Under $\tilde{\mathbb{P}}$, each outcome has probability $1/4$, so:

$$\begin{aligned} \tilde{\mathbb{P}}\{X = 1\} &= \frac{1}{2}, \quad \tilde{\mathbb{P}}\{S_1 = 8\} = \frac{1}{2}, \\ \tilde{\mathbb{P}}\{X = 1, S_1 = 8\} &= \tilde{\mathbb{P}}\{\text{HT}\} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}. \end{aligned}$$

One can verify all four nontrivial combinations factor similarly, so $\sigma(X) \perp \sigma(S_1)$ under $\tilde{\mathbb{P}}$.

- (iv) Under $\mathbb{P}$: $\mathbb{P}\{X = 1\} = \mathbb{P}\{\text{HT}\} + \mathbb{P}\{\text{TH}\} = \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$, $\mathbb{P}\{S_1 = 8\} = \frac{4}{9} + \frac{2}{9} = \frac{2}{3}$, but $\mathbb{P}\{X = 1, S_1 = 8\} = \mathbb{P}\{\text{HT}\} = \frac{2}{9}$. Since $\frac{4}{9} \cdot \frac{2}{3} = \frac{8}{27} \neq \frac{2}{9}$, independence fails under $\mathbb{P}$.

- (v) Under $\mathbb{P}$, the probability of heads is $2/3$. If $X = 1$, then $S_2 = 4$, which occurs at either HT or TH. Being told $X = 1$ rules out both HH (requiring two heads) and TT (requiring two tails), leaving only the mixed outcomes. This shifts probability mass toward the event $\{S_1 = 2\}$ (the first toss was T), so the posterior distribution of S_1 given $X = 1$ differs from the marginal: indeed $\mathbb{P}\{S_1 = 8|X = 1\} = \mathbb{P}\{\text{HT}\}/\mathbb{P}\{X = 1\} = \frac{2/9}{4/9} = \frac{1}{2} \neq \frac{2}{3} = \mathbb{P}\{S_1 = 8\}$.
-

Exercise 2.3 (Rotating the Axes). Let X and Y be independent standard normal ran-

dom variables. Let θ be a constant, and define

$$V = X \cos \theta + Y \sin \theta \quad \text{and} \quad W = -X \sin \theta + Y \cos \theta.$$

Show that V and W are independent standard normal random variables.

Solution

Since X and Y are jointly normal (being independent normals), any linear combination is also normal. V has mean 0 and variance $\cos^2 \theta + \sin^2 \theta = 1$, so $V \sim N(0, 1)$. Similarly $W \sim N(0, 1)$. Their covariance is

$$\begin{aligned} \text{Cov}(V, W) &= \mathbb{E}[VW] = \mathbb{E}[(X \cos \theta + Y \sin \theta)(-X \sin \theta + Y \cos \theta)] \\ &= -\cos \theta \sin \theta \mathbb{E}[X^2] + \cos^2 \theta \mathbb{E}[XY] + (-\sin^2 \theta) \mathbb{E}[XY] + \sin \theta \cos \theta \mathbb{E}[Y^2] \\ &= -\cos \theta \sin \theta + 0 + 0 + \sin \theta \cos \theta = 0, \end{aligned}$$

using $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = 1$ and $\mathbb{E}[XY] = 0$ (independence). Since (V, W) is jointly normal and $\text{Cov}(V, W) = 0$, V and W are independent.

Exercise 2.4. Let X be standard normal and Z independent with $\mathbb{P}\{Z = 1\} = \mathbb{P}\{Z = -1\} = \frac{1}{2}$. Define $Y = XZ$.

- (i) Establish the joint MGF formula $\mathbb{E}[e^{uX+vY}] = e^{\frac{1}{2}(u^2+v^2)} \cdot \frac{e^{uv}+e^{-uv}}{2}$.
- (ii) Show that $\mathbb{E}[e^{vY}] = e^{v^2/2}$, so $Y \sim N(0, 1)$.
- (iii) Use (i) and Theorem 2.2.7(iv) to show that X and Y are not independent.

Solution

- (i) Conditioning on Z :

$$\mathbb{E}[e^{uX+vY}] = \mathbb{E}[e^{uX+vXZ}] = \frac{1}{2} \mathbb{E}[e^{(u+v)X}] + \frac{1}{2} \mathbb{E}[e^{(u-v)X}] = \frac{1}{2} e^{\frac{1}{2}(u+v)^2} + \frac{1}{2} e^{\frac{1}{2}(u-v)^2}.$$

Factoring: $e^{\frac{1}{2}(u+v)^2} = e^{\frac{1}{2}(u^2+v^2)+uv}$ and $e^{\frac{1}{2}(u-v)^2} = e^{\frac{1}{2}(u^2+v^2)-uv}$, so

$$\mathbb{E}[e^{uX+vY}] = e^{\frac{1}{2}(u^2+v^2)} \cdot \frac{e^{uv}+e^{-uv}}{2}.$$

- (ii) Setting $u = 0$: $\mathbb{E}[e^{vY}] = e^{\frac{1}{2}v^2} \cdot \frac{e^0+e^0}{2} = e^{v^2/2}$, the MGF of $N(0, 1)$.
- (iii) $\mathbb{E}[e^{uX}] \cdot \mathbb{E}[e^{vY}] = e^{u^2/2} \cdot e^{v^2/2} = e^{\frac{1}{2}(u^2+v^2)}$. However, $\mathbb{E}[e^{uX+vY}] = e^{\frac{1}{2}(u^2+v^2)} \cosh(uv)$, which equals $e^{\frac{1}{2}(u^2+v^2)}$ only when $\cosh(uv) = 1$, i.e., when $uv = 0$. For $u, v \neq 0$ the joint MGF does not factor, so X and Y are not independent by Theorem 2.2.7(iv).

Exercise 2.5. Let (X, Y) have joint density

$$f_{X,Y}(x, y) = \begin{cases} \frac{2|x|+y}{\sqrt{2\pi}} \exp\left\{-\frac{(2|x|+y)^2}{2}\right\} & \text{if } y \geq -|x|, \\ 0 & \text{otherwise.} \end{cases}$$

Show that X and Y are standard normal random variables and that they are uncorrelated but not independent.

Solution

Marginal of X . For $x \geq 0$, substituting $u = 2x + y$ (so $du = dy$, and $y \geq -x$ corresponds

to $u \geq x$:

$$f_X(x) = \int_{-x}^{\infty} \frac{2x+y}{\sqrt{2\pi}} e^{-(2x+y)^2/2} dy = \int_x^{\infty} \frac{u}{\sqrt{2\pi}} e^{-u^2/2} du = \left[-\frac{1}{\sqrt{2\pi}} e^{-u^2/2} \right]_x^{\infty} = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

By the symmetry $f_{X,Y}(x,y) = f_{X,Y}(-x,y)$, the same holds for $x < 0$, so $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $X \sim N(0,1)$.

Marginal of Y. For any $y \in \mathbb{R}$, using $|x|$ and symmetry in x :

$$f_Y(y) = 2 \int_0^{\infty} \frac{2x+y}{\sqrt{2\pi}} e^{-(2x+y)^2/2} dx.$$

Substitute $u = 2x + y$, $du = 2 dx$:

$$f_Y(y) = \int_y^{\infty} \frac{u}{\sqrt{2\pi}} e^{-u^2/2} du = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}.$$

So $Y \sim N(0,1)$.

Zero correlation. $\mathbb{E}[XY] = \iint xy f_{X,Y}(x,y) dx dy$. Since $f_{X,Y}(x,y) = f_{X,Y}(-x,y)$ and xy is odd in x , the integrand is odd in x and the integral is 0. Hence $\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$.

Dependence. The joint density does not factor as $f_X(x) \cdot f_Y(y)$ since $f_{X,Y}(0,-1) = 0$ but $f_X(0)f_Y(-1) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{2\pi}} e^{-1/2} > 0$. By Theorem 2.2.7(v), X and Y are not independent.

Exercise 2.6. Consider a probability space $\Omega = \{a, b, c, d\}$ with $\mathcal{F}$ the full power set. Define a probability measure by $\mathbb{P}\{a\} = \mathbb{P}\{b\} = \mathbb{P}\{c\} = \mathbb{P}\{d\} = \frac{1}{4}$. Define random variables

$$X = \mathbb{I}_{\{a,b\}} \quad \text{and} \quad Y = \mathbb{I}_{\{a,c\}}.$$

- (i) Find $\sigma(X)$ and $\sigma(Y)$.
- (ii) Show that X and Y are independent.
- (iii) Compute $\mathbb{E}[X|Y]$ directly from Definition 2.3.1.

Solution

- (i) $\sigma(X) = \{\emptyset, \{a, b\}, \{c, d\}, \Omega\}$ and $\sigma(Y) = \{\emptyset, \{a, c\}, \{b, d\}, \Omega\}$.
- (ii) The joint probabilities all equal $\frac{1}{4}$, and the marginals are $\mathbb{P}\{X = 1\} = \mathbb{P}\{Y = 1\} = \frac{1}{2}$. One checks that $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for all $A \in \sigma(X)$, $B \in \sigma(Y)$; for example, $\mathbb{P}\{X = 1, Y = 1\} = \mathbb{P}\{a\} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$.
- (iii) We must find an $\sigma(Y)$ -measurable function Z satisfying $\int_A Z d\mathbb{P} = \int_A X d\mathbb{P}$ for all $A \in \sigma(Y)$. On $\{a, c\}$: $\int_{\{a,c\}} X d\mathbb{P} = \mathbb{P}\{a\} = \frac{1}{4}$, so Z must satisfy $Z(a) \cdot \frac{1}{4} + Z(c) \cdot \frac{1}{4} = \frac{1}{4}$. On $\{b, d\}$: $\int_{\{b,d\}} X d\mathbb{P} = \mathbb{P}\{b\} = \frac{1}{4}$, so $Z(b) \cdot \frac{1}{4} + Z(d) \cdot \frac{1}{4} = \frac{1}{4}$. Since Z must be constant on each atom of $\sigma(Y)$ (i.e., constant on $\{a, c\}$ and on $\{b, d\}$), we get $Z = \frac{1}{2}$ on both atoms. Therefore $\mathbb{E}[X|Y] = \frac{1}{2} = \mathbb{E}[X]$, consistent with independence.